



Maanav Nakodkar <maanav0114@gmail.com>

High School Student with Machine Learning Experience

17 messages

Maanav Nakodkar <maanav0114@gmail.com>
To: Joseph.Lautenbach@dnr.ohio.gov

Thu, Mar 26, 2026 at 12:21 PM

Hi Dr. Lautenbach,

I was told to reach out to you by Mrs. Nethero regarding how I can work with you using my Machine Learning/Computer Science knowledge. I'm sure you've already been told about my research accomplishments, but I've attached my paper anyway in case you haven't. I'm sure you know that it's not a proper publication yet as the second phase decisions are not out yet. Regardless, I can apply my skills to any domain that deals with data; I did so with astronomy/astrophysics in my paper, so I'm curious to see how we can work together regarding biology—if we can create an arrangement to do so.

I'm curious to know what you think. If you need any other information from me regarding previous experience or other projects, please let me know.

Thanks,
Maanav Nakodkar

 **IJCAI_Exoplanet_Paper.pdf**
320K

Maanav Nakodkar <maanav0114@gmail.com>
To: Joseph.Lautenbach@dnr.ohio.gov

Mon, Mar 30, 2026 at 9:09 AM

Dear Mr. Lautenbach,

I sent an email last Friday and am yet to hear back from you. I'm sending this as a followup to make sure you received it.

Thanks,
Maanav N.
[Quoted text hidden]

Joseph.Lautenbach@dnr.ohio.gov <Joseph.Lautenbach@dnr.ohio.gov>
To: Maanav Nakodkar <maanav0114@gmail.com>

Tue, Mar 31, 2026 at 4:02 PM

Hi Maanav,

Thanks for reaching out, and apologies for the slow response. I was out in the field collecting data over the past few days.

I've heard a bit about your work, and I agree that there are some exciting opportunities to apply machine learning to biology and natural resources. A few ideas that come to mind:

- Using large community-science datasets like iNaturalist to identify areas where box turtles may face elevated road-mortality risk.
- Predicting high-priority areas to survey for rare or declining species based on habitat, land-use, or environmental data.
- Identifying flood-prone agricultural lands that may be strong candidates for restoration (such as wetlands [hydric soils] or grasslands [non-hydric soils in floodplains]). This could help highlight places where habitat restoration aligns with variable agricultural productivity.

If none of these ideas feel like the right fit, I'd be happy to brainstorm additional possibilities. There's a lot of potential for machine learning to help reduce road mortality, guide targeted surveys, and focus conservation programs where they can be most effective.

Thanks again,
Joseph



Joseph Lautenbach
Wildlife Biologist
Ohio Division of Wildlife
Office: 614-265-6367 / Cell: 614-307-1992
Email: Joseph.Lautenbach@dnr.ohio.gov
wildohio.gov



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From: Maanav Nakodkar <maanav0114@gmail.com>
Sent: Monday, March 30, 2026 9:09 AM
To: Lautenbach, Joseph <Joseph.Lautenbach@dnr.ohio.gov>
Subject: Re: High School Student with Machine Learning Experience

You don't often get email from maanav0114@gmail.com. [Learn why this is important](#)

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Maanav Nakodkar <maanav0114@gmail.com>
To: "Joseph.Lautenbach@dnr.ohio.gov" <Joseph.Lautenbach@dnr.ohio.gov>

Tue, Mar 31, 2026 at 5:42 PM

Mr. Lautenbach,

Thanks for getting back to me! To answer your question, I think I would primarily be interested in problems that tackle guiding surveys. The second project you mentioned fits this perfectly, but I'm open to brainstorming more ideas if there's something that's more impactful or of higher priority in this category. I also had a question regarding how this arrangement would work. If you could clarify what you had in mind, then that would be appreciated.

Thanks,
Maanav N.

[Quoted text hidden]

Maanav Nakodkar <maanav0114@gmail.com>
To: "Joseph.Lautenbach@dnr.ohio.gov" <Joseph.Lautenbach@dnr.ohio.gov>

Thu, Apr 2, 2026 at 8:00 AM

Hi Mr. Lautenbach,

I'm sending a quick follow up since you did not see my email from Tuesday.

Thanks,
Maanav N.

[Quoted text hidden]

Joseph.Lautenbach@dnr.ohio.gov <Joseph.Lautenbach@dnr.ohio.gov>
To: Maanav Nakodkar <maanav0114@gmail.com>

Thu, Apr 2, 2026 at 3:19 PM

Hi Maanav,

That sounds great. Let me reach out to a colleague to see if there are any particular species with a high priority that would be of interest to her.

As far as the arrangement goes, I assumed that you were volunteering. Please let me know if you had something else in mind or if you had different expectations.

Thanks,
Joseph



Joseph Lautenbach
Wildlife Biologist
Ohio Division of Wildlife
Office: 614-265-6367 / Cell: 614-307-1992
Email: Joseph.Lautenbach@dnr.ohio.gov
wildohio.gov



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From: Maanav Nakodkar <maanav0114@gmail.com>

Sent: Thursday, April 2, 2026 8:00 AM

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Maanav Nakodkar <maanav0114@gmail.com>
To: "Joseph.Lautenbach@dnr.ohio.gov" <Joseph.Lautenbach@dnr.ohio.gov>

Thu, Apr 2, 2026 at 3:39 PM

Thanks for getting back to me.

When I first reached out to Mrs. Nethero to find an opportunity, it was in the hopes that I would be able to get a paid position. If it is possible to arrange this, then please let me know. Regardless, I also wanted to know whether I am needed to come in person somewhere to work or not.

Thanks for the help,
Maanav N

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18 attachments



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Joseph.Lautenbach@dnr.ohio.gov <Joseph.Lautenbach@dnr.ohio.gov>
To: Maanav Nakodkar <maanav0114@gmail.com>

Fri, Apr 3, 2026 at 12:06 PM

Hi Maanav,

Unfortunately, I cannot pay you. I don't have the resources available at this time. I completely understand if this is something that you no longer have an interest in. If you are still interested in volunteering, you would need to fill out a volunteer form (if you're over 18) or have a parent or guardian complete the volunteer form.

Let me know what you are thinking.

Joseph



Joseph Lautenbach
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From: Maanav Nakodkar <maanav0114@gmail.com>

Sent: Thursday, April 2, 2026 3:39 PM

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Maanav Nakodkar <maanav0114@gmail.com>

Fri, Apr 3, 2026 at 2:47 PM

To: "Joseph.Lautenbach@dnr.ohio.gov" <Joseph.Lautenbach@dnr.ohio.gov>

Hi Mr. Lautenbach,

I understand. I'm still willing to volunteer. Please let me know what I have to do next.

Maanav

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Maanav Nakodkar <maanav0114@gmail.com>

Thu, Apr 9, 2026 at 2:08 PM

To: "Joseph.Lautenbach@dnr.ohio.gov" <Joseph.Lautenbach@dnr.ohio.gov>

Good afternoon Mr. Lautenbach,

I'm following up on an email I sent last week. Please get back to me when you can.

Thanks,

Maanav

[Quoted text hidden]

Maanav Nakodkar <maanav0114@gmail.com>

Tue, Apr 14, 2026 at 8:00 AM

To: "Joseph.Lautenbach@dnr.ohio.gov" <Joseph.Lautenbach@dnr.ohio.gov>

Hi Mr. Lautenbach,

I'm still interested in the opportunity. I'm emailing since you haven't responded to my last couple emails. Please let me know what I have to do to next regarding this opportunity.

Thanks,

Maanav N

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Joseph.Lautenbach@dnr.ohio.gov <Joseph.Lautenbach@dnr.ohio.gov>

Wed, Apr 15, 2026 at 6:53 AM

To: Maanav Nakodkar <maanav0114@gmail.com>

Hi Manaav,

Sorry for the slow responses. This has been an extremely busy spring for me, with lots of field work.

I reached out to another biologist. We would be interested in trying to predict potential survey locations for salamanders in the Ambystoma genus. Have you heard of iNaturalist? This is a website/app where people can report observations of plants and animals. This would probably be the best resource to search for where these salamanders have been reported.

Perhaps you could explore this website and how to download data to start? I would search for Ambystoma in and outside of Ohio.

Thanks,

Joseph

From: Maanav Nakodkar <maanav0114@gmail.com>

Sent: Tuesday, April 14, 2026 8:00:00 AM

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Maanav Nakodkar <maanav0114@gmail.com>

Fri, Apr 17, 2026 at 8:00 AM

To: "Joseph.Lautenbach@dnr.ohio.gov" <Joseph.Lautenbach@dnr.ohio.gov>

Hi Mr. Lautenbach,

Thanks for getting back to me. I understand you're busy so there's no rush. I'll take a look at iNaturalist when I get some time this weekend.

I have a question about how the model would be used so that I have a better idea of what to look for. My current understanding is that you all wish to identify salamanders of the *Ambystoma* genus and want to know which places would have the highest probability of finding them. In other words, given a location (not necessarily lat. and long. but rather the attributes of a location that may affect salamanders like soil type etc.), the model could output a score that represents the environment's suitability for hosting the *Ambystoma*. A low score doesn't mean that there are little or no salamanders there, it just means that the environment isn't as suitable for *Ambystoma* which subsequently *may* mean that there are little *Ambystoma* present; Vice versa for a high score. Could you confirm if this is what the purpose would be or clarify what it would be if this isn't it?

Thanks,
Maanav N.

[Quoted text hidden]

Maanav Nakodkar <maanav0114@gmail.com>

Mon, Apr 20, 2026 at 10:52 AM

To: "Joseph.Lautenbach@dnr.ohio.gov" <Joseph.Lautenbach@dnr.ohio.gov>

Hi Mr. Lautenbach,

I'm sending a followup for my email from last week.

Thanks,
Maanav N.

[Quoted text hidden]

Maanav Nakodkar <maanav0114@gmail.com>

Fri, Apr 24, 2026 at 8:00 AM

To: "Joseph.Lautenbach@dnr.ohio.gov" <Joseph.Lautenbach@dnr.ohio.gov>

Hi Mr. Lautenbach,

I did familiarize myself a bit with the data on iNaturalist, but I still need to know the end goal to determine if it's feasible. Please refer back to my last email for more information.

Thanks,
Maanav N.

[Quoted text hidden]

Joseph.Lautenbach@dnr.ohio.gov <Joseph.Lautenbach@dnr.ohio.gov>

Mon, Apr 27, 2026 at 10:01 AM

To: Maanav Nakodkar <maanav0114@gmail.com>

Hi Maanav,

Sorry for the slow reply. We have some field intensive projects that have kept me away from my desk.

I was thinking of something similar to what you described. I think it might be possible to use known locations of *Ambystoma* salamanders and appropriate covariates (soil type, forest cover, etc.) to predict other locations to conduct surveys for salamanders in that genus. If I were to do this assessment, I would be using some sort of presence-only model (like a MaxEnt model) or a random forest model to predict potential locations. A brief review of the literature shows that there are a few newer models that I would need to familiarize myself with that show promise. I fully understand that salamanders could be present in areas that are predicted to have low suitability for the critter of interest. I think a score to predict likelihood of presence would be incredibly helpful. I typically try to stay away from categorical classifications and prefer to use a continuous metric, where possible. Does this answer your question?

I should be in the office a lot more this week and will be able to reply to emails much more promptly.

Take care,
Joseph



Joseph Lautenbach
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From: Maanav Nakodkar <maanav0114@gmail.com>

Sent: Monday, April 20, 2026 10:52 AM

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Maanav Nakodkar <maanav0114@gmail.com>

Mon, Apr 27, 2026 at 11:17 AM

To: "Joseph.Lautenbach@dnr.ohio.gov" <Joseph.Lautenbach@dnr.ohio.gov>

Mr. Lautenbach,

Thanks for getting back to me. I believe that this does answer my question. If I have any more questions, I'll be sure to let you know.

Thanks,
Maanav

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